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(Key and Solutions for AIMCAT2422)

Key

SECTION – I

1. D
2. A
3. B
4. D
5. D
6. C
7. B
8. A
9. B
10. A
11. D
12. A
13. B
14. D
15. A
16. D
17. 1432
18. B
19. 4312
20. B
21. B
22. B
23. C
24. 3124

SECTION – II

1. B
2. 760
3. C
4. 360
5. A
6. A
7. D
8. 200
9. 500
10. D
11. A
12. C
13. 110
14. B
15. B
16. C
17. B
18. C
19. D
20. A

SECTION – III

1. A
2. 66
3. C
4. D
5. B
6. D
7. A
8. A
9. B
10. 42
11. C
12. 13
13. 60
14. 7957
15. 3
16. 6
17. D
18. C
19. A
20. D
21. 28
22. B

Solutions

SECTION – I

Solutions for questions 1 to 4:

Number of words and Explanatory notes for RC:

Number of words: 569

1. The idea flow of the passage is:

Para 1 – Ideology, the way Hegel looks at it, could be secular, political, religious, cultural, moral/ social.

Para 2 – Hegel does not see a distinction between the ideological world and the material (practical/ temporal) world. What is true in one world could well have an impact on the other and vice-versa.

Para 3 – What underlies everything (i.e., both worlds) is consciousness.

Para 4 – Marx (consequently, the Left) has an entirely contradictory view. He equates consciousness with ideologies, and opines that they are the result of experience/ circumstances in the material (practical/ temporal) world.

Para 5 – This view is also held by the capitalist Right.

Para 6 – Weber's view (which turns out to be similar to Hegel's) is that ideals drive actions in the material world, and to understand that relationship, we have to study where it all begins – the spirit (i.e., consciousness).

Hence we can say that the roots of all ideology, according to the Hegelian view, lie in consciousness. Choice D is the answer.
Refer to para 1. Ideology in this sense is not restricted to the secular and explicit political doctrines we usually associate with the term, but can include religion, culture, and the complex of moral values underlying any society as well. This thought is somewhat reiterated in para 3. ***This consciousness may not be explicit and self-aware, as are modern political doctrines, but may rather take the form of religion or simple cultural or moral habits***. But this does not imply that the roots of all ideology lie in what has been given in choices (A), (B) and (C). The roots of ideology can be traced to consciousness (according to para 3).

Choice (D)

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2. Refer to para 5. The materialist bias of modern thought is characteristic not only of people on the Left who may be sympathetic to Marxism, but of many passionate anti-Marxists as well. Indeed, there is on the Right what one might label the Wall Street Journal school of deterministic materialism that discounts the importance of ideology and culture. This makes choice A the correct answer. The remaining choices do not apply. Choice (A)

3. Option A: Choice A is contradicted by the passage which emphasizes the doctrine of idealism. In fact, according to the passage, consciousness acts beyond the purview of the material world, the history of humanity and the future of a nation.
Option B: The passage introduces Hegel's perspective on consciousness and the role that it plays in determining various factors such as human history and the material world. The realm of consciousness in the long run necessarily becomes manifest in the material world, indeed creates the material world in its own image. To understand the emergence of modern capitalism and the profit motive one had to study their antecedents in the realm of the spirit. Hence choice B is the answer.
Option C: Though it has been mentioned that Hegel's view was that the real world didn't conform, choice C is not the primary concern of the passage.
Option D: Consciousness exhibits itself by taking the form of the religious, cultural and moral inclinations in a given society. This is one of the premises in the passage. But the passage goes much more beyond this aspect to explain how consciousness determines all human behaviour in the material world. The word 'proves' also renders this choice wrong. Choice (B)

4. Option A: The Wall Street Journal school of deterministic materialism discounts the importance of ideology and culture and ***sees man as essentially a rational, profit-maximizing individual***. Hence choice A is true and is not the answer.
Option B: This can be inferred from '***This consciousness may not be explicit and self-aware***, as are modern political doctrines, but may rather take the form of religion or simple cultural or moral habits.' So, there might not be complete awareness. So choice B is correct and is not the answer.
Option C: Paul Kennedy ascribed the decline of great powers to simple economic overextension. Therefore choice C is true and is not the answer.
Option D: Choice D is not correct and is the answer. Hegel ***did not believe*** that … ***the "material" world could not impinge on the ideal***. Hence, this is the correct answer.
Choice (D)

Solutions for questions 5 to 8:

Number of words and Explanatory notes for RC:

Number of words: 532

5. According to Margulis, the coming together of two species to form a new symbiotic consortium, which she called a holobiont. According to Rohwer, a holobiont is a host and all of its associated microorganisms.
Option A: Margulis's holobiont does not include a host. It is also not mentioned the Rohwer's holobiont is a community of microorganisms that reproduce by sending out tiny cells. Hence, this is not the answer.
Option B: Margulis's holobiont does not talk about a host. This is not true. Hence, this is not the answer.
Option C: Rohwer's holobiont has a host and its associated microorganisms. Although in the example of corals, polyps reproduce by cloning themselves, there is no mention in the passage that it is the same case with every Rohwer holobiont. Therefore, this cannot be inferred to be true. Hence, this is not the answer.
Option D: Both of this are in tune with the definitions given by Margulis and Rohwer. Hence, this is the answer. Choice (D)

6. The author mentions the phrase, "an enemy of an enemy is a friend, after all", in the fourth paragraph of the passage, where the author talks about coral systems in order to explain Rohwer's definition of a holobiont. In this context, "an enemy" is the virus, while "an enemy of an enemy" refers to the corals, because the corals produce mucus which kills the virus.
Option A: Although this can be attributed to being a defense mechanism of the holobiont because both the corals and the useful bacteria help each other in surviving, the above phrase does not establish the same. The phrase only states that the symbiotic relationship helps them fight mutual threats and not the idea that it is a defense mechanism. Hence, this is not the answer.
Option B: The statement highlights why the bacteria find associating with the corals beneficial. Since the corals kill the viruses, the bacteria treats the coral as a "friend". It is not to illustrate how the corals benefit from the bacteria (this is established in the statement "Corals don't possess …"). Hence, this is not the answer.
Option C: In this phrase, bacteria can reason that an enemy of an enemy, i.e., the corals, are their friends because the corals kill the virus harmful to the bacteria. Hence, this is the answer.
Option D: The phrase is not about the corals being of benefit to the virus. The passage does not mention any symbiotic relation between the virus and the corals. Hence, this cannot be the answer.
Choice (C)

7. The author discusses corals in the fourth paragraph of the passage.
Option A: The author talks about Margulis's definition of 'holobiont', wherein there are two organisms, and then goes on to say in the fourth paragraph that two protagonists just aren't enough when it comes to explaining the evolutionary success of corals. From this, it can be inferred that

Margulis definition does not explain the evolutionary success of corals. Therefore, this is definitely true. Hence, this is not the answer.

Option B: In the fourth paragraph, it is mentioned that the most spectacular corals work hand-in-hand with photosynthetic algae that they host. From the passage, we can infer that the most spectacular corals have photosynthetic algae but there is not enough information to infer that most corals have photosynthetic algae. Therefore, it cannot be said that this option is definitely true. Hence, this is the answer.

Option C: In the fourth paragraph, it is mentioned that it is the coral polyps that reproduce by cloning themselves. Therefore, this is definitely true. Hence, this is not the answer.

Option D: It is mentioned in the fourth paragraph that corals do not have a complicated immune system to fend off pathogens and that they need to selectively cultivate helpful or benign bacteria, which crowd out the harmful microbes. From this, it can be inferred that this is definitely true. Hence, this is not the answer. Choice (B)

8. *"A major implication of this theory is that natural selection doesn't just act on the genome of individual organisms: it acts on the hologenome of holobionts, which are seen as single units that can evolve at the level of the holobiont."*

Option A: It is stated that evolutionary forces act on the hologenome of holobionts. In the last paragraph, it is stated that the hologenome is made up of the host genome and all the microbial genomes. Hence, this is the answer.

Option B: It is stated that natural selection acts on the hologenome, which includes the genome of individual microorganisms. Therefore, we cannot say that it does not act on the genome of individual organisms. Hence, this is not the answer.

Option C: The passage does not talk about the similarities between a host genome and its hologenome and how natural selection will be affected by it. Hence, this is not the answer.

Option D: The passage states that natural selection does not just act on the genome of individual organisms but also on the hologenome of the holobiont. It does not talk about natural selection being more important in the evolution of one species than in that of another. Hence, this is not the answer. Choice (A)

Solutions for questions 9 to 12:

degree, regulate conditions within their own bodies. Hence choice A is correct and is not the answer.

Option B: James Lovelock, who first proposed the Gaia theory, used an analogy with living organisms, all of which can, to some degree, regulate conditions within their own bodies. The theory brings to mind New Age crankery, and ***some of the research Mr Flannery cites to underpin it is speculative at best***…But having shown the reader a glimpse of this fascinating byway, he speeds straight past, impatient to reach the next intellectual stop, ***while the tentative theory is simply accepted***. Hence choice B is inconsistent with the information given in the passage. Choice B is the answer.

Option C: The "Gaia" theory is the subject of the opening chapters of the book. Hence choice C is true and is not the answer.

Option D: There is an effort to organise the chapters around two competing models of human behaviour, ***a co-operative, far-sighted wisdom*** that Mr Flannery dubs the "Gaian" approach. .... Hence choice D is true and is not the answer.

Choice (B)

10. Statement (a): "Here on Earth" a twin biography, of humanity and the planet it inhabits, but that description is inadequate. Mr Flannery's subject is ***the likely fate of humankind, and whether the powers granted to modern civilisation by science and technology will prove to be its downfall or its salvation.*** Hence (a) is true and is not the answer.

Statement (b): (b) is negated by the first para. His earlier book "The Weather Makers" was a best seller. (His 2005 book, "The Weather Makers", about climate science and global warming, was a bestseller.)

Statement (c): The book "is still worth reading, though less for answers than for its interesting hypotheses." Hence, (c) is correct and is not the answer.

Statement (d): (d) is true from the last sentence of para 2. There is an effort to organise the chapters around ***two competing models of human behaviour*** .....

Statement (e): Statement (e) cannot be inferred from the passage. In fact, by telling us that he easily accepts certain hypotheses, the passage indicates that he's quite willing, in some instances, to take a side.

Number of words and Explanatory notes for RC:

Number of words: 504

9. Option A: The "Gaia" theory, which is the subject of the opening chapters, starts with the surprising observation that such things as the Earth's surface temperature and the salinity of its oceans have fluctuated remarkably little over billions of years despite, for instance, a significant increase in the power of the sun. James Lovelock, who first proposed the Gaia theory, used an analogy with ***living organisms, all of which can, to some***

So statements (b) and (e) are not true.
Choice (A)

11. Statement (a): Mr Flannery is a respected biologist with plenty of published papers to his name (para 3, second sentence). He is also a mammalogist. Tim Flannery is a paleontologist (para 1, first sentence). A mammologist is one skilled in the study of mammals. A paleontologist is a scientist who specializes in the study of life forms that existed in previous geologic periods, as represented by their fossils: Hence statement (a) is true and is not the answer.

Statement (b): The book feels dilettantish, with a dizzying array of concepts introduced, briefly discussed, then dispensed with before the reader has had time to digest them. ..... Fewer examples more deeply explored might have added up to a more convincing case, and an easier read. .... Having shown the reader a glimpse of this fascinating byway, he speeds straight past, impatient to reach the next intellectual stop, while the tentative theory is simply accepted. But statement (b) would be correct if it said 'cursory' or 'superficial'. 'Rambling' would be more in the nature of 'directionless'. But he's not really directionless, it's just that he doesn't stop long enough at points that need further examination. **So, as it is,** (b) is not true and is the answer.
Statement (c): Statement (c) is supported by the last sentence of para 1. Mr Flannery's subject is the likely fate of humankind, and whether the powers granted to modern civilisation by science and technology will prove to be its downfall or its salvation. So statement (c) is true and is not the answer.
Statement (d): There is an effort to organise the chapters around two competing models of human behaviour, a co-operative, far-sighted wisdom that Mr Flannery dubs the "Gaian" approach and a reckless, ultimately destructive short-termism that he calls "Medean", after the bloodthirsty enchantress of Greek myth. But Statement (d) (unsure about the validity of the Gaian and Medean approaches) cannot be inferred from the passage.
So, statements (b) and (d) are not true.
Choice (D)

Solutions for questions 13 to 16:

Number of words and Explanatory notes for RC:

Number of words: 538

13. Refer to the fifth para.
Option A: Choice A is not true. Craftsmen and pattern-books did come over from the Protestant Low Countries.
Option B: The Renaissance was generally slow to arrive in England, ***largely because Elizabeth's troublesome relations with Catholic Europe made the free exchange of ideas difficult.*** Our relative isolation from the European cultural mainstream led to a national style which was a bizarre though attractive mixture of Gothic and classical styles. Hence choice B is true.
Option C: The passage does not talk about the archaic and stringent rules of architecture in Elizabeth's reign. Hence choice C is incorrect.
Option D: The underlying ethos of medieval architecture was 'fitness for purpose'. The buildings of the 16th century were also governed by fitness for purpose. There was a building boom in the 16th century. However, choice D is not specific to the question.
Choice (B)

14. "However, the truth is not as simple as **that**" is the sentence (the first sentence of the second para) which has to be considered for this question. The sentence ("not as simple as **that**") actually makes the reader pause and review the previous sentence and what follows later. (A misconception is a view or opinion that is incorrect because it is based on faulty thinking or understanding.) A

12. Option A: Refer to the third para and the first two lines of para 4. Choice A can be inferred. Refer also to the last sentence of para 4. But having shown the reader a glimpse of this fascinating byway, he speeds straight past, impatient to reach the next intellectual stop, while the tentative theory is simply accepted. From this, it can be inferred that the book deals perfunctorily (cursorily) with a wide range of topics. Hence choice A is the answer.
Option B: The chapters of the book center around two competing models of human behaviour. But choice B (odds with each other) cannot be inferred. Choice B is not a limitation.
Option C: The word 'inability' in choice C renders it incorrect (He muses on whether humanity counts as a superorganism (a classification usually reserved for bees and ants), why we have yet to discover intelligent aliens). Choice C is not a limitation of the book.
Option D: Fewer examples more deeply explored might have made for a more convincing case, and an easier read. The book has many illustrations, but the concepts have not been explained in detail. So, choice D is distorted.

Choice (A)

misconception in the first para (It used to be thought that once the Romans pulled out of Britain in the fifth century ..... fell into decay as British culture was plunged into the Dark Ages) has been corrected in the second para (Romano-British culture including architecture survived long after the Roman withdrawal.) Hence choice D is true.
Option A: The word 'hypothesis' in choice A renders it incorrect. Also, there are no arguments presented in the first sentence of the second para and the following lines.
Option B: There is no 'theory' that has been challenged here. Hence choice B is totally incorrect.
Option C: Choice C is not true. There is no reiteration of a point. The choice fails to capture the contrasting viewpoint that the following sentence puts forward. A contrast word 'however' has been used in the first sentence of the third para.

Choice (D)

15. Option A: ***A greater sense of security led to more outward-looking buildings***, as opposed to the medieval arrangement where the need for defence created houses that faced inward onto a courtyard or series of courtyards. ***This allowed for much more in the way of exterior ornament***. Hence choice A is correct.

Option B: The underlying ethos of medieval architecture was 'fitness for purpose'. The buildings of the 16th century were also governed by fitness for purpose. So choice B is not correct.
Option C: Choice C cannot be inferred. In the fifth paragraph we are told: Relative isolation of England from the European cultural mainstream led to a national style which was a bizarre though attractive mixture of Gothic and classical styles. The author considers the same 'attractive' and not 'dreary'.
Option D: Choice D does not relate to the 16th century. Architectural ideas could be borrowed from the French, the Dutch and the Italians during the 1640s to 1650s (the time of the Civil War).

Choice (A)

So, 143 is a block ('more sensible' is also like an answer to the question raised in 4). 2 offers the conclusion about the ***real*** radicalism in poetry – referencing the question in Sentence 1 about only 'work that pushes boundaries'. So, **1432**.

Ans: (1432)

Solution for question 18:

18. Option A: This option misses the important point of research needed in order to combine the benefits of the three parameters. Also, the third parameter of regulating meal size has not been mentioned. So, the option effectively misses the essence of the para that research needs to be directed towards integrating the three parameters to develop a regimen that can help with aging related chronic diseases.

16. Refer to the seventh and eight paragraphs.

Option A: There was a passion for the Baroque grandeur which Louis XIV had turned into an instrument of statecraft at Versailles. So choice A can be inferred.

Option B: The British Baroque was a reassertion of authority, an expression of absolutist ideology by men who remembered a world turned upside down during the Civil War. Hence choice B can be inferred.

Option C: Formal, geometrical and symmetrical planning meant that a great lord could sit in his dining chamber .... radiating out in straight lines to either side. His gardens would reflect those lines in long, straight walks and avenues. The style is heavy and rich, sometimes overblown and melodramatic (sensational). So choice C can also be inferred.

Option D: Nothing about Baroque being a culmination of other styles has been mentioned in the end of the passage. Hence, Option D is the answer. Choice (D)

Solution for question 17:

17. Sentence 1 is an independent idea about the wrong or troubling belief that only work that pushes boundaries is worth reading. It can be noticed that Sentence 4 offers an example to this idea, where poems pushing boundaries are compared to others. Same goes for Sentence 3.

Sentence 2 starts with a 'perhaps' and offers a summary of sorts – 'the real radicalism in poetry lies in acknowledging and valuing more conservative forms' and this needs context.

Sentence 3 has a comparison – 'More sensible' with examples like Alice Oswald's poems (so basically, the second part of the sentence doesn't add value). The clue is 'these poems' referring to a group from elsewhere.

Sentence 4 raises a rhetorical question – 'Can one definitively say they (poems such as Spelt) are better than poems where the conventions of the form reasserted themselves?' This sentence is grammatically independent, but we need a context.

So, the independent idea 1 comes first, followed by 4, since 3 has 'these' referencing poems in 4.

Option B: This option is closest to the essence of the para as it depicts the two ideas in the para – that there should be an effort to combine the three factors of balanced diet, fasting and controlled meal size, and that such a regimen should be used to tackle the burden of geriatric (age-related) diseases. Option B is the answer.

Option C: The research according to the para should be about 'integration of a balanced nutritious diet with controlled meal size and patterns and periods of fasting' so that better strategies can be developed to reduce the burden of diseases proliferating with age. Hence, the research itself has been inaccurately depicted in this option. Also, the factor of meal size hasn't been mentioned here. Only two factors – good diet and fasting have been mentioned. Hence, Option C is not the answer.

Option D: The main focus of the para is not what older people can or cannot do. Also, it cannot be understood from the para that older people can prevent the onset of diseases through diet control and fasting. The author states that research needs to be done so this goal of preventing age-related diseases can be achieved. Hence, Option D is not the answer. Choice (B)

Solution for question 19:

19. Sentence 1 starts with a contrast marker 'But' followed by a sentence with a negative tone talking about light from many sources blocking fireflies pairing up.

Sentence 2 refers to fireflies in a previous sentence as 'them' and it has a negative tone (that light pollution is outshining their mating signals).

Sentence 3 is a grammatically independent sentence that talks about how males use light signals to attract females.

Sentence 4 talks about how fireflies use their glowing power to attract their partners (introduction of a concept).

3 is downstream to 4 since, 4 is more generic, and introduces the concept, while 3 elaborates on it with specifics. Also, males of what species is understood only from 4. So, 43 is a block. 1 and 2 give us the bad news. But, 1 starts with 'But', so, 2 cannot precede it. So, the order is 4312.

Ans: (4312)

Solution for question 20:

Solution for question 23:

20. The sentence cannot fit in blank (1) because the first two sentences provided in the para form a block, the 'these followers' in the second sentence referring to the avid movie-goers.

It has to be blank (2) because Dudeism has been introduced in the preceding sentence, followed by 'the followers of Dudeism'. These two can be separated by a sentence about the start of the cult following which is upstream to what the followers of Dudeism practice.

It cannot be blank (3) because the preceding sentence is about how Lebowski has influenced culture, and we have already moved away from the followers. Similarly, it cannot be the concluding sentence of the para, especially with the sentence preceding blank (4) already being a conclusive statement.

Choice (B)

Solutions for questions 21 and 22:

21. The author mentions the commonly held belief, puts forth the claims by behaviourists and explains it using an example. This claim is best captured by choice (B). Choice (A) states that there is nothing called mind, while the behaviourists cast doubts about states of minds. Also the term "neuron" is unnecessarily introduced. Choice (C) is incorrect as it states that both body and mind contribute significantly to a man's actions and this cannot be drawn from the passage. While Choice (D) is correct according to behaviourists, it is not a general view. The para mentions that traditionally belief is considered to be a state of mind. But behaviourists do not consider it so. This part is missing in this option and, hence, is not the correct answer.

Choice (B)

22. Option A: Instead of the essence of the para, this option zeroes down to examples mentioned in the para. It also doesn't give us what a small world network really is. Hence, Option A is not the answer.

Option B: This option best describes the essence of the para which talks about what a small world network is – a group of clusters which are connected to each other but not in a systematic way, necessarily. Hence, Option B is the answer.

Option C: The word 'organised' completely misses the point and makes this summary highly inaccurate. Small world networks are somewhere between random and clustered networks. Clustered networks aren't necessarily organised. Hence, Option C is easy to eliminate.

Option D: This option is inaccurate given that it doesn't mention that there are connections between these clusters. That is what differentiates small world networks from clustered networks. Hence, Option D is not the answer.

Choice (B)

23. It cannot be blank (1) because the promise in the next sentence refers to the promise in the starting sentence of the para – 'promised a world of applications.'

It cannot be blank (2) because the next sentence 'And…' continues the same line of thought as the first sentence of the para, the clue being 'incredibly durable materials.' So, we cannot jump to graphene and then again come back to these materials.

It has to be blank (3) because we talk about the date before its properties were identified during discovery and research. For the same reason, it cannot be blank (4) because how graphene was first discovered should precede its properties, rather than following it.

Choice (C)

Solution for question 24:

24. Sentence 1 talks about someone travelling to the western coast of Africa to observe the longest solar eclipse. This sentence is obviously dependent on another which tells us who 'his' is.

Sentence 2 talks about Eddington's plans. We can understand that this sentence has to follow another sentence that introduces Eddington, as this sentence despite being grammatically independent, is contextually dependent.

Sentence 3 introduces the mission – what was Eddington trying to do. He is setting out to test Einstein's theory. (Note that we don't know what is the ingenious experiment, which means that such a sentence would be downstream to this.)

Sentence 4 is a dependent sentence since it is preceded by something that gives us the context of Hyades and what Eddington planned to observe.

1 is downstream to 3 as it gives us an idea of what experiment nature has furnished – total solar eclipse. 1 is upstream to 2, since its idea (solar eclipse) is bigger than the elaboration of what Eddington was planning to observe (the light of the Hyades).

4 is downstream to 2 because it tells us the end conclusion, or the expected result, of the whole mission. This sentence cannot occur unless the setup and logic has already been explained. So, the order is 3124.

Ans: (3124)

Difficulty level wise summary - Section I	
Level of Difficulty	Questions
Very Easy	–
Easy	21
Medium	9, 10, 12, 13, 15, 16, 19, 22, 23, 24
Difficult	1, 2, 3, 4, 5, 7, 11, 14, 18, 20
Very Difficult	6, 8, 17

SECTION – II

Solutions for questions 1 to 5:

1. From the graph, we can see that Kalyan was eliminated in the first lap. Kalyan would have been eliminated when Jitesh finished his first lap (since the average speed of Jitesh is the lowest among the remaining four students).
   Time taken for Jitesh to finish the lap = 400/2 = 200 seconds.
   Hence, Kalyan would have been in the race for 200 seconds and would have covered 200 × 1.8 = 360m. Choice (B)

2. We can see that Jitesh was eliminated in the second lap. Jitesh would have been eliminated when the third person finished the second lap.
   Time taken by Gaurav, Himesh and Inder to finish the first lap will be 120 seconds (400/3.33), 160 seconds (400/2.5) and 150 seconds (400/2.67) respectively.
   Time taken by them to finish the second lap will be 180 seconds (400/2.22), 120 seconds (400/3.33) and 160 seconds (400/2.5) respectively.
   Total time taken to finish the first two laps by Gaurav, Himesh and Inder will be 300 seconds, 280 seconds and 310 seconds respectively.
   Hence, Jitesh would be eliminated after 310 seconds of which he spent 200 seconds in the first lap. In the next 110 seconds, he would have covered 359.7 meters.
   The total distance covered by Jitesh in the race would be 400 + 360 = 760 m. Ans: (760)

3. The fastest to finish the first two laps of the race was Himesh (in 280 seconds). Choice (C)

4. The runner up of the race would be eliminated when the winner finishes his final lap.
   Time taken by Gaurav to finish the first two laps are 120 seconds and 180 seconds.
   Time taken by Gaurav to finish the third lap = 400/1.67 = 240 seconds.
   Time taken by Himesh to finish the first two laps are 160 seconds and 120 seconds.
   Time taken by Himesh to finish the third lap = 400/2 = 200 seconds
   Total time taken by Gaurav to finish the first 3 laps = 540 seconds.
   Total time taken by Himesh to finish the first 3 laps = 480 seconds.
   If Gaurav won the race, the time taken by Gaurav to finish the last lap would be 400/3 = 133.33 seconds. Hence, Himesh would have run for 60 seconds (the difference between the time taken by the two in the first three laps) and 133.33 seconds i.e., a total of 193.33 seconds.
   In 193.33 seconds, Himesh would have covered 2.22 × 193.33 = 429 meters. Clearly, this is not possible.
   Therefore, Himesh must have won the race. He would have finished the final lap in 400/2.22 = 180 seconds.
   Gaurav would have run for 180 – 60 = 120 seconds in the fourth lap covering a distance of 3 × 120 = 360 m. Ans: (360)

5. The total duration of the race = the time taken by Himesh to finish the race = 160 + 120 + 200 + 180 = 660 seconds. Choice (A)

Solutions for questions 6 to 10:

The maximum time that any person could have spent at the supermarket is 5 hours, i.e, 300 minutes.

6. Since each person spent at least ₹100, we can take four persons to have spent ₹100 each and the fifth person to have spent the remaining 700 – 400 = ₹300.
   If a person spent ₹300, he could not have spent more than 150 minutes in the supermarket.
   Hence, the maximum possible time that any person could have spent in the supermarket is 150 minutes. Choice (A)

7. Given that Pavan spent more than Lohit and also more than Kalyan. Since four of the five persons spent the same amount, Pavan must have been the one who spent a different amount.
   If the amount spent by Pavan is P and the amount spent by Lohit or Kalyan is L, then P = L + 200 = 2L.
   Hence, amount spent by Lohit (and all the others expect Pavan) is 200 and the amount spent by Pavan is 400.
   The minimum time that Pavan can be in the supermarket is $\frac{400}{3} = 133.33$ minutes.
   Choice (D)

8. Since the amount and time that a person spent are correlated, to maximize the amount that Hari spent, we can maximize the time that Hari spent.
   From the given ratio, the minimum amounts that Kalyan, Lohit and Ratan spent are 100, 200 and 300 respectively.
   The minimum durations (in minutes) that they spent in the supermarket must be 33.33, 66.67 and 100 minutes.
   Given that at most one person was in the supermarket at any given time. Of the total 300 minutes available, these three persons would have been in the supermarket for 33.33 + 66.67 + 100 = 200 minutes.
   Of the remaining 100 minutes, Pavan must be in the supermarket for a minimum of 33.33 minutes (assuming he spent ₹100).
   Hence, Hari could have spent a maximum of 66.67 minutes in the supermarket.
   The maximum amount (in ₹) that Hari could have spent in the supermarket is 66.67 × 3 = 200.
   Ans: (200)

9. Assuming that four persons spent ₹100 each in the supermarket and stayed in the supermarket for 33.33 minutes, the fifth person could have been in the supermarket for 300 – (33.33 × 4) = 166.67 minutes.
   The maximum amount that can be spent by the fifth person is 166.67 × 3 = ₹500. Ans: (500)

10. We can start off by assuming that all the five persons were in the supermarket for 300 minutes. The amounts that they spent must be between 600 and 900. Also, each person spent a distinct amount, and each person spent a multiple of 100. Since there are only four distinct multiples of 100 between 600 and 900, only four persons could have been in the supermarket for 300 minutes. If the fifth person spent ₹500, then he can be in the supermarket for a maximum of 250 minutes. Hence, all the five persons could have been in the supermarket for a maximum of 250 minutes.

Choice (D)

Solutions for questions 11 to 15:

The score of Drew in Test 3 is not given. But among his top four scores, the scores of Test 1, Test 2 and Test 5 will definitely be considered (if he scored high marks in Test 3, that score will be considered in Applicable Score; if he did not, then the score in Test 6 will be considered).

For John, among the four top scores, Test 1, Test 2 and Test 4 will definitely be considered. The sum of these three scores is 233. Since the Applicable Score is 321, John must have scored 321 – 233 = 88 marks in one of the tests. Hence, John must have scored 88 marks in Test 5. For John, the scores in Test 1, Test 2, Test 4 and Test 5 will be considered.

For Kevin, the scores in Test 1, Test 2 and Test 6 will definitely be considered.

For Phil, the scores of Test 3, Test 4, Test 5 and Test 6 are given. These four scores cannot be the top 4 scores as the sum of these scores does not add up to 256. If three of these four scores are considered, then the scores of Test 3, Test 4 and Test 5 will be considered. The sum of these three scores is 155. Hence, Phil must have received 101 marks in the other test which is not possible (as each test is for 100 marks). Hence, only two of the three scores must be considered. Hence, Phil must have scored a total of 256 – 75 – 65 = 116 in the other two tests (Test 1 and Test 2). The scores of Test 1, Test 2, Test 3 and Test 4 will be considered for Phil.

For Paul, the top 3 of the given scores add up to 213. He must have scored 54 marks in the other test (his score in Test 1). Hence, for Paul, the scores of Test 1, Test 2, Test 3 and Test 5 will be considered.

For Jack, the top 3 of the given scores add up to 172 marks. He must have scored 68 in Test 3. Hence, for Jack, Test 2, Test 3, Test 5 and Test 6 will be considered.

For Gary, the scores of Test 1, Test 2 and Test 4 will be considered. The other score can be of Test 3 or Test 6. However, the score of Test 3 cannot be considered because for Phil, the scores of these four tests were considered. Hence, for Gary, the score of Test 6 would have been considered. Hence, the scores of Test 1, Test 2, Test 4 and Test 6 must have been considered for Gary and the Applicable Score of Gary is 264.

For Drew, the scores in Test 1, Test 2 and Test 5 are considered. The other score that can be considered is Test 3 or Test 6. However, Test 3 cannot be considered because then the same set of four tests will be considered for Paul and Drew. Hence, for Drew, the scores of Test 1, Test 2, Test 5 and Test 6 would have been considered.

For Kevin, the scores in Test 1, Test 2 and Test 6 are considered. The other score cannot be of Test 5 (as Drew's Applicable Score considers the same set of tests) and it cannot be Test 4 (as Gary's Applicable Score considers the same set of tests). Hence, Test 3 must be considered for Kevin. The Applicable Score of Kevin must be 263.

The following table provides the scores of each student in each test, with the cells containing the four highest scores of each student highlighted:

Student	Test 1	Test 2	Test 3	Test 4	Test 5	Test 6	Applicable Score
Drew	**52**	**48**	<45	33	**66**	**45**	211
John	**92**	**70**	54	**71**	**88**	23	321
Kevin	**65**	**70**	**54**	<54	47	**74**	263
Phil	**x**	**116 – x**	**75**	**65**	15	14	256
Paul	**54**	**78**	**77**	<54	**58**	46	267
Jack	41	**49**	**68**	44	**75**	**48**	240
Gary	**62**	**75**	<56	**71**	38	**56**	264

11. The Applicable Score of Gary is 264.
Choice (A)

12. Kevin has the fourth highest Applicable Score.
Choice (C)

13. The difference between the highest and the lowest Applicable Score is 321 – 211 = 110.
Ans: (110)

14. The scores of Test 4 were considered for the least number of students, i.e., three students.
Choice (B)

15. The total marks scored by all the seven students can be determined accurately for two tests – Test 5 and Test 6.
Choice (B)

Solutions for questions 16 to 20:

The total distance that Raghu has to travel to visit the eight cities, A through H, are 400, 600, 500, 300, 200, 800, 700 and 400, respectively.

16. Given that he has to visit at least one city among C, E, G and H. The order of these four cities, from nearest to farthest, are E (100 km), H (200 km), C (250 km) and G (350 km).
Also, he has to visit at least two cities among A, B, D and F. The order of these four cities, from

We need to remove two cities among A, D, E and H from this list. If we remove the two cities that are the farthest (i.e., A and H) and replace them with two remaining cities which are nearest (i.e., B and G), the five cities will be E, D, C, B and G. In this case, the total distance exceeds 1000 km. Hence, any combination of five cities will not be possible. Hence, he can visit a maximum of 4 cities. This can be done in multiple ways, a one of which is (E, A, B, F).
Choice (B)

18. From the previous solution, the maximum number of cities that he can visit is 4.
Since he has to visit at least two cities from A, B, D and F, we can start with this.
If he visits A, B, he has to visit E. This is a possibility with a total distance of 1200 km.
Further, he can visit A, B, E and one of F or G, for a distance of 2000 km or 1900 km.
If he visits A and D, he has to visit E. This is not possible as this violates (iii).
If he visits A and F, he has to visit E. This is one possibility with a distance of 1400 km. He cannot add any more cities to this combination of cities.
If he visits B and D, he can visit G alone (1600 km) or H alone (1300 km) or G and H together (2000 km). These are three possibilities.
If he visits B and F, he cannot visit G (as it exceeds 2000 km) but can visit H for a total distance of 1800 km.

nearest to farthest, are D (150 km), A (200 km), B (300 km) and F (400 km).

Since he has to visit at least three cities, he can choose the three cities that are nearest (with one from the first group and two from the second group).

These three are E, D and A. However, this will violate condition (iii).

If he chooses H instead of E, it will still violate condition (iii).

If he chooses E, he has to choose A (from (v)). Instead of D he can visit B. Hence, he can visit E, B and A, travelling a total of (100 + 300 + 200) * 2 = 1200 km.

He can also choose H, D and B. In this case, however, he will travel longer, i.e., for a total of (200 + 300 + 150) × 2 = 1300 km

Hence, the minimum distance that he must travel is 1200 km. Choice (C)

17. Given that Raghu cannot travel for more than 2000 km. The sum of the distances of the cities that he visits (from Metropolis) must be at most 1000 km.

From (iii), we know that he cannot visit two cities out of A, D, E and H. That leaves a maximum of 6 cities.

From (vi), he cannot visit one of B or C. That leaves a maximum possible value of 5 cities.

Suppose he visits five cities. The five cities which are nearest are E, D, A, H and C. The total distance in this case is 900 km (one way).

However, this violates condition (iii).

If he visits D and F, he can visit G or H as the third city (for 1800 km and 1500 km respectively).

Hence, the possibilities are (A, B, E), (A, B, E, F), (A, B, E, G), (A, F, E), (B, D, G), (B, D, H), (B, D, G, H), (B, F, H), (D, F, G), (D, F, H).

In any of the possible cases, he would not have visited C. Choice (C)

19. From the possible cases, in the above solution, Raghu must have selected (D, F, H). (The total distance in this case is 300 + 800 + 400 = 1500 km.) Hence, he must have visited H. Choice (D)

20. If Raghu visits B, then he has to visit D. From the previous solution, the route with the least distance which includes both B and D is (B, D, H) for 1300 km. If he does not visit B, he has to travel at least 1400 km, for (A, F, E). Hence, the minimum possible distance that Raghu can travel is 1300 km. Choice (A)

Difficulty level wise summary - Section II

Level of Difficulty	Questions
Very Easy	–
Easy	1, 6
Medium	2, 3, 7, 9, 10, 11, 12, 13, 14, 15, 16, 17
Difficult	4, 5, 8, 18, 19, 20
Very Difficult	–

SECTION – III

Solutions for questions 1 to 22:

1. Remainder of 4^N, where N is any natural number, when divided by 6, is always 4. Choice (A)

2. Let the ages of Aditi and her mother be denoted by *ab* and *ba* respectively.

Their ages, before 15 years were $(10a + b - 15)$ and $(10b + a - 15)$

It is given that $(10a + b - 15) = \dfrac{(10b + a - 15)}{3}$

$\Rightarrow 3(10a + b - 15) = (10b + a - 15)$

$\Rightarrow 29a = 7b + 30$

5. $30\% = \frac{3}{10}$

$20\% = \frac{1}{5}$

	Cost Price	Selling Price
30% loss	10	7
20% profit	5	6

Since selling price is the same,

	Cost Price	Selling Price
30% loss	60	42
20% profit	35	42
	95	84

∴ Together, he bought both articles at ₹95 and sold at ₹84.

Since a and b are both single digit numbers, only $a = 2$ and $b = 4$ is a possible solution.

$\therefore$ Aditi's age is ab i.e., 24 and her mother's age is ba. i.e., 42.

The sum of their present ages is 66.

Ans: (66)

3. We have $-2 < a < 1$, $-3 < b < -1/3$, $-3 < c < -1/3$, $d = ab/c$.

b/c (the ratio of the two negative quantities) is always positive. Its minimum value is $(1/3)/3 = 1/9$ and its maximum value is $3/(1/3) = 9$

$\therefore 1/9 < b/c < 9$ and $-2 < a < 1$

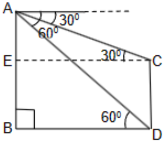
As $\frac{b}{c} < 9$ and $a > -2$, $\frac{ab}{c} > -18$

As $\frac{b}{c} < 9$ and $a < 1$, $\frac{ab}{c} < 9$

$\therefore -18 < \frac{ab}{c} < 9$

Only choice (C) is necessarily true. Choice (C)

4.



Let us denote the tower and the building by AB and CD respectively.

It is given that BD = EC = 40

In Δ AEC, $\tan 30° = \frac{AE}{EC}$

$$AE = \frac{40}{\sqrt{3}}\left[\because \tan 30 = \frac{1}{\sqrt{3}}\right]$$

In Δ ABD, $\tan 60° = \frac{AB}{BD}$

$$\Rightarrow AB = 40\sqrt{3}\left[\because \tan 60 = \sqrt{3}\right]$$

Height of the building CD = AB – AE

$= 40\sqrt{3} - \frac{40}{\sqrt{3}} = \frac{80}{\sqrt{3}}$ m

Choice (D)

$\therefore$ Loss % = $\frac{11}{95} \times 100 = 11.58\%$ Choice (B)

6. $(57)_8 = (5 \times 8) + 7 = (47)_{10}$

As 47 is not a perfect square $(57)_8$ cannot be a perfect square. The remaining numbers are perfect squares. Choice (D)

7. The weights are in A.P. If we take an odd number of terms of an AP, the average of all the terms is also a term of the A.P. But for an even number of terms, it is not. As two of the AWs are the equal to the weights of the stones in the two bags, those two bags should contain an odd number of stones and the other bag, an even number of stones.

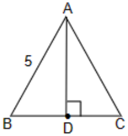
A	$2k+1$	$2k-1$
B	$2k+1$	$2k-1$
C	$2k$	$2k$

$\therefore n = 6k + 2$ or $6k - 2$

Hence, it can be 82, but not 99, 90 or 113.

Choice (A)

8.



Let AB = 5

and BC = 10

Draw a perpendicular from A to BC

Now $\frac{1}{2} \times BC \times AD$ = Area of $\Delta ABC = 20 = \frac{1}{2} \times 10 \times AD$

$\Rightarrow AD = 4$

$\therefore BD = \sqrt{AB^2 - AD^2}$

$= \sqrt{5^2 - 4^2} = 3$

$\therefore DC = BC - BD = 7$

$\therefore$ In Δ ADC

$AC = \sqrt{AD^2 + DC^2}$

$= \sqrt{4^2 + 7^2} = \sqrt{65}$ Choice (A)

Note: In this case ∠ACB is an acute angle. Say θ. These conditions will also be satisfied when ∠ACB = 180° – θ. But BC will not be the longest side.

9. The difference between a three-digit number and its reverse is always a multiple of 99.
Let the number be $8k$, then its reverse is $3k$.
$\therefore$ The difference = $5k = 99m$
$\Rightarrow k$ is multiple of 99 and m is a multiple of 5.
$\therefore$ The difference is a multiple of 5×99,
i.e. 495, 990 ………
Since the difference between a number and its reverse is at most 891, the difference has to be 495.
$\Rightarrow k = 99$
$\therefore$ The number if 8×99
i.e. 792
So, only one such number is possible.
Choice (B)

10. Let the age of the teacher be T years.
Total age of the students = 59×22
Total age of the class with the teacher = $59 \times 22 + T$
Total age of the class with the teacher = $60 \times 22\frac{1}{3}$

$59 \times 22 + T = 60 \times \frac{67}{3}$

$\therefore$ T = 1340 − 1298 = 42 years. Ans: (42)

11. $g(x) = \min(7 - 3x, 3 + 5x)$
The two expressions $7 - 3x$ and $3 + 5x$ are equal when $x = \frac{1}{2}$

For $x < \frac{1}{2}$, $7 - 3x > 3 + 5x$

$\therefore g(x) = 3 + 5x < 5\frac{1}{2}$

For $x = \frac{1}{2}$, $7 - 3x = 3 + 5x$

$\therefore g(x) = 7 - 3x = 3 + 5x = \frac{11}{2}$

For $x > \frac{1}{2}$, $7 - 3x < 3 + 5x$

$\therefore g(x) = 7 - 3x < 5\frac{1}{2}$

$\therefore$ The maximum value of $g(x)$ is $5\frac{1}{2}$ or $\frac{11}{2}$.
Choice (C)

12. Let the number of ₹2 and ₹5 coins used be denoted by x and y respectively
Now $2x + 5y = 120$. Where $x, y \geq 0$
The solutions are

x	60	55	50	…	0
y	0	2	4	…	24

Thus, there are 13 ways in all. Ans: (13)

13. Let the speed at which the escalator is descending be e steps per second.
Let the time taken by A be t seconds.
$30 + te$ = Total number of steps in the escalator.
In the time A covers 30 steps, B would cover 60 steps.
As B has covered 120 steps so the time taken must be $2t$.
$120 - 2te$ = Total number of steps in the escalator
$\therefore 30 + te = 120 - 2te$
$3te = 90$
$\Rightarrow te = 30$.
Thus the total number of steps visible on the escalator.
$= 30 + te = 60$ Ans: (60)

14. $f(25) = 25^3 - 25^2 - f(24)$
$= 25^3 - 25^2 - 24^3 + 24^2 + f(23)$
$= 25^3 - 25^2 - 24^3 + 24^2 + 23^3 - 23^2 - f(22)$
$= (25^3 - 24^3 + 23^3 - 22^3 +.- 2^3) - (25^2 - 24^2 + 23^2 - 22^2 + \ldots. + 2^2) + f(1)$
Now, $(25^3 - 24^3 + 23^3 - 22^3 + \ldots. + 3^3 - 2^3)$
$= (1^3 + 2^3 + 3^3 + \ldots. + 25^3) - 2(2^3 + 4^3 + 6^3 + \ldots + 24^3) - 1^3$

$= \left(\frac{25 \times 26}{2}\right)^2 - 16(1^3 + 2^3 + \ldots + 12^3) - 1$

$= (325)^2 - 16\left(\frac{12 \times 13}{2}\right)^2 - 1$

$= 8280$
Similarly, $25^2 - 24^2 + 23^2 - 22^2 + \ldots + 3^2 - 2^2$
$= (1^2 + 2^2 + \ldots. + 25^2) - 2(2^2 + 4^2 + \ldots + 24^2) - 1$
$= \frac{25 \times 26 \times 51}{6} - 8(1^2 + 2^2 + \ldots + 12^2) - 1$

$= 25 \times 13 \times 17 - 8\left(\frac{12 \times 13 \times 25}{6}\right) - 1$

$= 324$.
$\therefore$ f (25) = 8280 – 324 + 1 = 7957 Ans: (7957)

15. Given,
$xy + x + y = 215$ ………(1)
$yz + y + z = 161$ ………(2)
$zx + z + x = 107$ ………(3)
where, x, y, z are positive integers.
From (1)
$xy + x + y + 1 = 216$
$\Rightarrow x(y + 1) + (y + 1) = 216$
$\Rightarrow (x + 1)(y + 1) = 216$ ………(4)
Similarly,
$(y + 1)(z + 1) = 162$ ………(5)
and $(z + 1)(x + 1) = 108$ ……….(6)
(4) (6) ÷ (5) gives

$$\frac{[(x+1)(y+1)][(z+1)(x+1)]}{[(y+1)(z+1)]} = \frac{216\,(108)}{162}$$

$$= \frac{12(18)(12)(9)}{18(9)}$$

$(x + 1)^2 = 12^2 \Rightarrow x + 1 = 12 \Rightarrow x = 11$
Similarly, $(y + 1)^2 = 18^2 \Rightarrow y + 1 = 18 \Rightarrow y = 17$
And $(z + 1)^2 = 9^2 \Rightarrow z + 1 = 9$
i.e., $(x, y, z) = (11, 17, 8)$
$\frac{y-z}{x-z} = \frac{17-8}{11-8} = \frac{9}{3} = 3$. Ans: (3)

16. As $\frac{p}{q} = \frac{q}{r} = \frac{r}{s} = 6$

$\frac{p}{q} = \frac{pq}{q^2} = 6$, $\frac{q}{r} = \frac{qr}{r^2} = 6$

$\frac{r}{s} = \frac{rs}{s^2} = 6$

Therefore $\frac{pq+qr+rs}{q^2+r^2+s^2} = 6$ Ans: (6)

17. Given $4x = 25 - 3y$

$\Rightarrow x = \frac{25-3y}{4}$

Now, substituting the value of x in the equation of the circle, we get

$\frac{(25-3y)^2}{4} + y^2 = 25$

$625 - 150y + 9y^2 + 16y^2 = 400$

$25y^2 - 150y + 225 = 0$

$y^2 - 6y + 9 = 0 \Rightarrow y = 3$

$\therefore x = \frac{25-3y}{4} = \frac{25-9}{4} = 4$

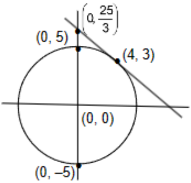
Therefore, the curves $x^2 + y^2 = 25$ and $4x + 3y = 25$ meet at exactly one point whose coordinates are (4, 3).

Again, the graphs of $4x + 3y = 25$ and $x = 0$ intersect at one point i.e., (y-intercept) at $\left(0, \frac{25}{3}\right)$ whereas $x^2 + y^2 = 25$ and $x = 0$ intersect at two points i.e., (0, 5) and (0, –5).

Hence, there are a total of four distinct points at which any two of the three given graphs meet.

Alternate solution:

By observation $4x + 3y = 25$, is a straight line and $x^2 + y^2 = 25$ is a circle with centre (0, 0). Now, the perpendicular distance of the centre (0, 0) from the straight line $4x + 3y - 25 = 0$ is $\frac{|4(0) + 3(0) - 25|}{\sqrt{4^2 + 3^2}} = 5$. Since this is equal to the radius of the circle, the straight line is tangent to the circle. Hence, we get a total of four distinct points (as explained in the above solution).



Choice (D)

18. Let the cost of the pen be p, and the cost of the book be b. Given $p + b = 53$.

Let the price of each pencil be n. Given that, $b = 6n + 21$

Also, $p = 2n \Rightarrow b = 3p + 21 \Rightarrow p = 8$.

Hence, he could have purchased 2 pens with ₹21.

Choice (C)

19. $|2y - 3| \le 11$

$\therefore -11 \le 2y - 3 \le 11$

$-11 + 3 \le 2y \le 11 + 3$

$-8 \le 2y \le 14$

$-4 \le y \le 7$. Choice (A)

20. The pulp content by weight in fresh grapes $= 100 - 80 = 20\%$

$\Rightarrow$ 10 kg of fresh grapes which have 2 kg of pulp, finally yield 2.5 kg of dry grapes.

This 2 kg of pulp is $\left(\frac{2}{2.5} \times 100\right)\%$ by weight of dry grapes.

$\Rightarrow$ Dry grapes have 80% pulp by weight.

Choice (D)

21. 20 men were employed to complete the work in say N days. Therefore, the estimated work is 20N mandays.

Work completed in $\frac{N}{3}$ days is 5N mandays.

The remaining time according to revised schedule $= \frac{3N}{4} - \frac{N}{3} = \frac{9N - 4N}{12} = \frac{5N}{12}$ Remaining work = 15N manday.

20 men in N/3 days do 5N mandays of work.

Number of men needed in 5N/12 days to do 15N mandays of work is $(20)\left(\frac{N}{3}\right)\left(\frac{12}{5N}\right)\left(\frac{15}{5}\right) = 48$

$\Rightarrow$ 28 additional men are needed.

Ans: (28)

22. In presence of so many notations, it is advisable to take small numbers and observe the pattern that emerges.

Since the label of the baskets goes on till 2n, let us assume values of n to begin with.

Say n = 1 $\Rightarrow$ The baskets are labelled 1, 2, 2 (since there are two baskets with label 2).

In this case, N = 3 (since there are three baskets in total)

and the total number of apples as per case I $= 2 + 4 + 4 = 10$

and the total number of apples as per case II $= 3 + 3 + 3 = 9$

$\therefore$ the difference when N = 3 is 1.

Hence, substituting N = 3 in the choices, choice (D) can be eliminated.

Similarly, when n = 2, the baskets are 1, 2, 2, 3, 3, 3, 4, 4, 4, 4.

$\therefore$ A total of 10 baskets are present.

$\therefore$ N = 10

Baskets with odd-numbered labels have one apple more in case II when compared with case I. In case of even-numbered baskets, it is vice-versa.

$\therefore$ Difference between total number of apples in case I and case II is the difference between the number of odd and even-numbered baskets.

For instance, difference between number of apples in basket 1 under case I and case II is +1.

For label 2, difference is –1.

For label 3, difference is +1.

.

.

.

and so on.

$\therefore$ For even-numbered basket, difference is –1 and for odd-numbered basket, difference is +1.

$\therefore$ Difference in total number of apples under case I and case II, for n = 2, is $|(1 + 3) - (2 + 4)| = 2$, i.e., when N = 10.

Now substituting N = 10 in the answer choices, choices (A) and (C) can be eliminated.

Hence, the correct answer must be choice (B).

Alternative Solution:

After the pattern in the difference is found as 'n', we get

$N = 1 + 2 + 3 + \ldots\ldots + (2n)$

$= \frac{2n(2n+1)}{2}$

$\Rightarrow 2n^2 + n - N = 0$

$\Rightarrow n = \frac{-1+\sqrt{1+8N}}{4}$ Choice (B)

Difficulty level wise summary - Section III	
Level of Difficulty	Questions
Very Easy	19
Easy	1, 5, 10, 12, 16, 18, 20
Medium	3, 4, 6, 8, 9, 21
Difficult	2, 7, 11, 13, 14, 15, 17, 22
Very Difficult	–

Done

Tip: Scroll slowly for the best results